Daniel Gause

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Education

B.A. Physics and Mathematics *September 2016 – May 2020*
Middlebury College

- GPA 3.5, *cum laude*
- Physics thesis: Structural Analysis of Potential Dual Quasar Pairs
- Mathematics thesis: An Introduction to Large Deviation Theory

Research and Professional Experience

Postbaccalaureate Researcher *September 2026 – Present*
National Radio Astronomy Observatory — Advisor: Adele Plunkett

- Compiling lists of nearby (<500 pc) Class 0/I protostars and Herbig–Haro objects from published catalogs
- Building a tested Python pipeline that harvests ALMA, MAST, NOIRLab, and other archive holdings and tags spectral-line coverage per target
- Identifying targets with multi-epoch observations suited to jet kinematics and proper motions

Research Intern *May 2026 – August 2026*
NSF-Simons AI Institute for Cosmic Origins (CosmicAI) — National Radio Astronomy Observatory

- Built MANNA, an open-source MCP server that lets AI assistants query astronomical data archives via IVOA standards; released on PyPI and the MCP Registry
- Assessed MCP performance across multiple LLMs with a custom evaluation framework; with MANNA, models ran up to 8.1× faster at up to 6.7× lower cost
- Deployed open-weight LLMs on NOIRLab Astro Data Lab GPUs and integrated the Astro Data Lab JupyterHub with jupyter-ai

Software Engineer *May 2021 – May 2026*
Accenture Federal Services

- AI/ML Engineer *June 2023 – May 2026*
  - Led development for two small teams building production GenAI services, including a FastAPI backend with Azure OpenAI and vector search
  - Built over a dozen Python packages with unit, integration, and functional tests and CI/CD
- Data Scientist *May 2021 – June 2023*
  - Trained survival models on >10M UNOS records (Apache Spark) to predict kidney-waitlist outcomes
  - Built ML models on NASA satellite imagery to predict wildfire, flood, and landslide risk

Research Assistant *May 2019 – March 2020*
Department of Physics, Middlebury College — Advisor: Eilat Glikman

- Cross-matched >1M known quasars with archival HST imaging; built a pipeline to reduce MAST data
- Ran surface-brightness contour analysis on thousands of images, identifying 21 dual-quasar candidates

Independent Research

Lead Developer, STIPS *December 2024 – Present*
University of California Observatories, Lick Observatory — with Kyle Westfall

- Lead developer of STIPS, adapting the LSST Science Pipelines to the Nickel and CTIO 1-m telescopes
- Calibrated Landolt standards to $\leq$12 mmag in $BVRI$; astrometry matches Gaia DR3 to 14–48 mas
- Reproduced published photometry of SNe 2023ixf and 2020wnt and recovered the HD 189733 b transit

Cataclysmic Variables Research Project *August 2024 – 2025*
City College of San Francisco — Advisor: Claia Bryja

- Observed cataclysmic variables, including AY Psc, in $BVRI$ on the Lick 1-m Nickel Telescope
- Wrote a Python reduction pipeline: bias, flat-field, astrometric, and photometric calibration

Publications

- Lichter-Marck, I., Powell, A. M., Wogan, K., **Gause, D.**, et al. (2026). "Perityleae B.G.Baldwin (Asteroideae): A catalogue of natural history, taxonomy, and morphology of the rock daisy tribe." *Capitulum*, 4(2), 31. doi:10.53875/capitulum.04.2.02
- **Gause, D.**, Nikutta, R., Mason, B., Plunkett, A., et al. "MANNA: an MCP Architecture for NOIRLab, NRAO, and additional Archives." *In preparation for AJ.*
- **Gause, D.** & Westfall, K. "STIPS: Bringing the LSST Science Pipelines to 1-Meter-Class Telescopes." *In preparation for PASP.*

Presentations

- Gause, D., Telkamp, Z., Plunkett, A., Mason, B., Nikutta, R., Juneau, S., Zacharia Anil, A., Loomis, R., Offner, S., Durrett, G., Torrey, P., & Murphy, E. "MANNA: MCP Architecture for NRAO and NOIRLab Archives." Poster, CosmicAI Cosmic Horizons Conference, Charlottesville, VA, July 2026.
- Gause, D. & Westfall, K. "The Small Telescope Image Processing Suite (STIPS): Bringing the LSST Science Pipelines to 1-m Telescopes." Poster, 248th Meeting of the American Astronomical Society, Pasadena, CA, June 2026.
- Gause, D. "On the Hunt for Dual Quasars." Keck Northeast Astronomy Consortium Symposium, Vassar College, October 2019.
- Gause, D. "Searching for DQSOs in Archival Hubble Data." Poster, Middlebury Summer Research Poster Session, July 2019.

Volunteer and Outreach

Scientific Computing Department Volunteer *March 2024 – September 2026*
California Academy of Sciences

- Built Python packages for herbarium data: DBSCAN clustering of specimens into collecting expeditions, and GBIF distribution maps for the rock daisy tribe (coauthor, *Capitulum* 2026)
- Parallelized a citation-matching app (40% faster); set up Apptainer under Slurm on the HPC cluster
- Trained models on EnMAP hyperspectral imagery to predict where tree species are absent

Service Learner Volunteer *August 2024 – January 2025*
City College of San Francisco

- Helped plan and host monthly public sky viewing nights and planetarium shows

Skills and Certifications

- **Languages:** Python, C++, C, SQL/ADQL, Bash
- **Astronomy:** LSST Science Pipelines, Gen3 Butler, Astropy, pyvo, FITS, IVOA protocols
- **Computing:** Slurm, HTCondor, Docker, Apptainer, HPC, CI/CD, pytest, Git
- **Certifications:** CompTIA Security+ (2025)